

PROPOSAL SUMMARY

PREPARED FOR

Mid-States Companies
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PROJECT

Eight (8) XM-4460 hammer mills for corn grinding — dry-grind ethanol plant
48–50 TPH per mill at 7,200 sq in screen area
Mills set on concrete; plenums hung below on the next floor
Air system (fans, filters, ducting) by W.D. Patterson — Mike King

REFERENCE

MCE-Q-2609-MIDSTATES-R1
Basis: Rough Rider hammer mill review of Sep 15, 2026 (C. Schaudt / J. Bliss / M. King) and MCE engineering of Sep 15, 2026
Validity: **through December 31, 2026** ·
Currency: USD
FOB MCE plant, Newkirk, OK

Midwest Custom Engineering is pleased to quote eight (8) XM-4460 extra-heavy-duty hammer mills with feeders, self-cleaning magnet adapters, 350 FPM plenum chambers and plenum transfer screws for corn grinding at the Mid-States ethanol project. Each mill is rated 48–50 TPH on corn with 7,200 sq in of screen area and a 500 HP drive. Toshiba 500 HP main drive motors are listed separately as item 6, with pricing to follow on receipt of the Toshiba quotation; Mid-States may instead purchase the motors directly and ship them to MCE for mounting. The air-relief system is by W.D. Patterson and is not included here.

1. DESIGN BASIS

per Jason Bliss, MCE, Sep 15, 2026

PARAMETER	DESIGN VALUE	NOTES
Product	Whole corn — dry-grind ethanol	Grinding for fermentation; screen selection per Mid-States process
Number of mills	8 × XM-4460	Extra-heavy-duty, direct-drive, dual grinding chamber, gravity discharge
Capacity	48–50 TPH per mill	Based on 7,200 sq in screen area per mill
Screen size	#5 (5/64")	Two (4 total) sets of screens per mill included
Mill drive	500 HP, 1800 RPM, 460 V/3/60, TEFC premium efficiency, 1.15 SF	Toshiba — priced separately as item 6 (Toshiba quote pending); may instead be supplied by Mid-States and shipped to MCE for mounting
Air relief through mill	9,360 CFM @ 6" SP per mill	Fans, dust filters, ducting and airlocks by W.D. Patterson
Feeder	14-row, 10" dia. nylon cup	20.72 RPM at 90% cup fill; auto self-cleaning magnet adapter
Plenum chamber	350 FPM design velocity, 18'-0" long	3/16" A36 plate, 2" flat bar wraps every 24"; hangs below mill floor — no support structure
Discharge	Transfer screw under each plenum	Sized for 50 TPH ground corn
Installation	Mills set on concrete; plenums on floor below	Installation by others

2. SCOPE OF SUPPLY AND PRICING

USD · taxes and freight excluded

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
1	XM-4460 Hammer Mill — Extra Heavy Duty Mill Design Direct drive, dual grinding chamber, gravity discharge hammer mill: • Motor base extension; reversible top feed inlet; screen carriage	8	\$123,791.80	\$990,334.40

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
	- Screen size: #5 (5/64") — two (4 total) sets of screens included - Construction: mild steel with AR liners 7,200 sq in screen area - Hammers: hard faced, heat treated — one (1) set included - HP range 300–600 HP; operating horsepower 500 HP - Capacity: 48–50 TPH corn; air relief 9,360 CFM @ 6" SP through the mill - Included: top feed inlet, mill/motor support base, three NEMA 4/12 limit switches, OSHA guard, vibration switch and rubber vibration pads - Motor: 500 HP, 1800 RPM, 460 V/3/60, TEFC premium efficiency, 1.15 SF (Toshiba, Baldor or Siemens) — not included in the mill price; see item 6. Customer-supplied motors are shipped to MCE for mounting and alignment before the mill ships.			
2	XM-10" AC 14-Row Nylon Cup Feeder with built-in Auto Self-Clean Magnet Adapter 10" dia. nylon cup rotor, 14 rows; shaft extension, motor base and roller chain drive with safety shield 1.608 cu ft/rev; 20.72 RPM at 90% cup fill - Combo unit built with auto self-clean magnet adapter: stainless steel front, stainless steel hinged magnet plate, air operated with two air cylinders and trash pans; SF300-50" plate magnet - Micro switch to indicate magnet door open - Gearhead motor quoted as item 3	8	\$41,631.70	\$333,053.60
3	Feeder Gearhead Motor 2 HP, 3 PH, 60 Hz, 460 V, TEFC gearhead motor, 40.7 RPM final output — sized for 20.72 RPM feeder speed through the roller chain drive.	8	\$4,059.00	\$32,472.00
4	Custom Plenum Chamber — 350 FPM, for XM-4460 18'-0" long; sized for 350 FPM design velocity - Constructed of 3/16" A36 plate, wrapped every 24" with 2" flat bar - Mounting flange for the hammer mill; plenum hangs below the mill floor (mill sets on concrete) — no support structure included - Discharge flange for transfer screw (item 5); mounting plate for dust filter or ducting flange as required - Exterior: one shop coat grey enamel - Plenum length may change with final mill placement; confirmed on approval drawings	8	\$44,567.60	\$356,540.80
5	Transfer Screw Conveyor — 18" dia. x 21'-0", for XM-4460 plenum 18" diameter screw conveyor, 21'-0" long inside trough ends, to convey 50 TPH of ground corn at 35 PCF, control fed at 45% trough loading, 46 RPM, horizontal. Length to be confirmed against the 18'-0" plenum on approval drawings. 18" RH sectional screw, 1/4" flights, full pitch, on 6" Sch 40 pipe, 3-7/16" shafts; Tem-U-Lac coupling bolts - Anti-friction tail trough end with WPS and lip seal; standard drive trough end with feet	8	\$32,862.50	\$262,900.00

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
	18" × 3/16" flared trough 18'-0" long, uncovered to act as inlet, with flared-to-U-trough adaptor plate; 3'-0" single-flanged trough with 12 GA gasketed cover 19" × 19" × 3/16" flush end discharge spout; support feet; shroud; flange gaskets and bolt kits - Drive: Baldor 10 HP, 1750 RPM, 3/60/460 V TEFC 215T motor; Dodge TA4207H25 taper-bushed reducer, class 2 gearing, 46 RPM output; drive shaft, motor mount, sheaves, belts and guard - Exterior: one shop coat grey enamel; general arrangement drawings for approval before fabrication			
6	Main Drive Motor — Toshiba 500 HP Toshiba 500 HP, 1800 RPM, 460 V/3/60, TEFC premium efficiency, 1.15 SF, 449T frame (Baldor or Siemens equivalent acceptable). Mounted, aligned and guarded on the mill at MCE. Priced separately on receipt of the Toshiba quotation; Mid-States may instead purchase the motors directly and ship them to MCE for mounting at no additional charge.	8	Pending Toshiba quote	Not in total
Subtotal — items 1 to 5				\$1,975,300.80
Less 10% discount project discount, 8-mill order				(\$197,530.08)
Total — eight (8) mill lines as quoted				\$1,777,770.72

Unit prices are quoted for a single order of eight (8) lines. The project discount applies to items 1–5 and will be applied to item 6 when the Toshiba motor price is added.

OPTIONS AND ADDERS

OPTION	DESCRIPTION	QTY	UNIT PRICE	NET ADDER (AFTER 10%)
A	Plenum chambers fabricated by Mid-States Mid-States may elect to fabricate the plenum chambers to MCE drawings. MCE furnishes fabrication drawings and the mill mounting flange details; items 1–3, 5 and 6 are unchanged. Confirm whether MCE supplies the mill mounting flange/adaptor with this option.			Deduct item 4

3. FURNISHED BY OTHERS

- Air-relief system for the mills — fans, dust control filters, ducting, airlocks and explosion protection — by W.D. Patterson (Mike King), sized to 9,360 CFM @ 6" SP per mill as stated above
- Mill drive motors if supplied by Mid-States (see item 6); motor starters, VFDs and motor control
- Controls, PLC programming and instrumentation beyond the limit switches, vibration switches and magnet micro switches listed
- Concrete, foundations, floor openings, anchor bolts and any support steel for the plenums
- Infeed to the feeders, and conveyance from the transfer screw discharge
- Compressed air to the magnet adapters (air cylinders)
- Installation, millwright, electrical and start-up labor

4. FREIGHT, DELIVERY AND SCHEDULE

ITEM	BASIS
Freight	FOB MCE plant, Newkirk, OK; freight to Mid-States site quoted at time of shipment or shipped freight collect.

ITEM	BASIS
Delivery	__ weeks after receipt of order, approval drawings and customer-supplied motors (if applicable). Order placement planned by Mid-States for December 2026; MCE will reserve fabrication capacity on receipt of a signed proposal.
Approval drawings	General arrangement drawings for the plenum chambers and transfer screws issued for Mid-States approval before fabrication; plenum length confirmed at that time.
Installation	By others. MCE start-up assistance and operator training available at MCE's standard daily rate plus expenses.

5. COMMERCIAL TERMS

Payment: 50% down with order, balance before shipment. Prices are in US dollars and exclude sales tax, freight and installation. Pricing is valid **through December 31, 2026**; steel and motor pricing are subject to adjustment for orders placed after that date. Warranty per MCE standard terms and conditions. This proposal covers the equipment listed in Section 2 only; the air system is quoted separately by W.D. Patterson.

PREPARED BY

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FOR MID-STATES COMPANIES

Accepted: _____

Name / Title: _____

Date: _____ PO #: _____